

Panel Data Econometrics

Lecture 1 Quiz: Foundations

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How to use this quiz

Coverage

These five questions cover the core logic of Lecture 1:

- why panel data change the comparison,
- why pooled OLS often fails,
- what fixed effects identify,
- how first differences compare with fixed effects,
- what two-way fixed effects absorb.

Deliberate omission

Inference and clustering are excluded here. This quiz is about identification, interpretation, and algebra.

Quiz 1

Question

Suppose we study the effect of worker training $Train_{it}$ on wages $Wage_{it}$ using panel data.

Why is the panel comparison often more credible than the pure cross-sectional comparison? State the key source of omitted-variable bias that panel data may help remove.

Solution 1

Answer

The cross-sectional comparison mixes workers with different permanent traits, such as ability, motivation, or firm quality. If those traits affect both training and wages, then

$$\text{Cov}(\text{Train}_{it}, \alpha_i) \neq 0,$$

where α_i is a time-invariant worker effect.

Interpretation

Panel data let us compare a worker to the worker's own past. If the omitted trait is stable over time, fixed effects or first differences can remove α_i and shift attention from across-worker comparisons to within-worker changes.

Quiz 2

Question

Consider the scalar panel model

$$y_{it} = \beta x_{it} + \alpha_i + u_{it}.$$

Show the probability limit of pooled OLS and explain in one sentence why pooled OLS can be inconsistent even when $\text{Cov}(x_{it}, u_{it}) = 0$.

Solution 2

Answer

The pooled OLS probability limit is

$$\text{plim } \hat{\beta}^{POLS} = \beta + \frac{\text{Cov}(x_{it}, \alpha_i)}{\text{Var}(x_{it})} + \frac{\text{Cov}(x_{it}, u_{it})}{\text{Var}(x_{it})}.$$

Interpretation

Even if x_{it} is uncorrelated with the idiosyncratic shock u_{it} , pooled OLS is still inconsistent whenever x_{it} is correlated with the permanent unit effect α_i . Then pooled OLS uses contaminated between-unit variation.

Quiz 3

Question

Start from

$$y_{it} = x'_{it}\beta + \alpha_i + u_{it}.$$

Write the within-transformed equation. Then answer: why can fixed effects not estimate the coefficient on a regressor that is constant over time within each unit?

Solution 3

Within transformation

Subtract the unit mean:

$$y_{it} - \bar{y}_i = (x_{it} - \bar{x}_i)' \beta + (u_{it} - \bar{u}_i).$$

Equivalently,

$$\ddot{y}_{it} = \ddot{x}_{it}' \beta + \ddot{u}_{it}.$$

Why time-invariant regressors drop out

If a regressor does not vary within unit, then $x_{it} = \bar{x}_i$ for all t , so

$$x_{it} - \bar{x}_i = 0.$$

Fixed effects therefore have no within-unit variation left from which to identify its coefficient.

Quiz 4

Question

In the model

$$y_{it} = x_{it}\beta + \alpha_i + u_{it},$$

compare fixed effects and first differences.

1. When do they produce exactly the same slope coefficient?
2. Why can first differencing be especially sensitive to measurement error?

Solution 4

Part 1

Fixed effects and first differences coincide when $T = 2$. With only two periods, demeaning and differencing extract the same single within-unit movement.

Part 2

If the observed regressor is

$$x_{it}^{obs} = x_{it}^* + \nu_{it},$$

then differencing gives

$$\Delta x_{it}^{obs} = \Delta x_{it}^* + (\nu_{it} - \nu_{i,t-1}).$$

So the transformed regressor contains two measurement-error draws, which can magnify attenuation bias.

Quiz 5

Question

Consider the two-way fixed-effects model

$$y_{it} = x'_{it}\beta + \alpha_i + \lambda_t + u_{it}.$$

Suppose one regressor can be written as

$$x_{it} = a_i + b_t.$$

Can its coefficient be identified under two-way fixed effects? Explain briefly.

Solution 5

Answer

No. Under two-way fixed effects, the identifying variation must survive both the unit sweep and the time sweep. But a regressor of the form

$$x_{it} = a_i + b_t$$

is completely absorbed by the unit effects and time effects.

Double-demeaning logic

After the transformation

$$\tilde{x}_{it} = x_{it} - \bar{x}_i - \bar{x}_t + \bar{x},$$

we get $\tilde{x}_{it} = 0$ for every (i, t) . So there is no residual variation left to estimate its coefficient.

Takeaways

1. Panel data improve credibility by shifting the comparison toward within-unit changes.
2. Pooled OLS often fails because it mixes within variation with contaminated between variation.
3. Fixed effects identify slopes from within-unit movements only.
4. First differences remove the same unit effect, but behave differently when $T > 2$ and can amplify measurement error.
5. Two-way fixed effects absorb both permanent unit differences and common period shocks.